

5.15 Classwork: Logarithms — Markscheme

1. [Maximum mark: 12]

(a) Calculate the value of each of the following logarithms. [6]

(i) $\log_4 \frac{1}{64}$

$$\frac{1}{64} = 4^{-3} \quad \text{M1}$$

$$\log_4 \frac{1}{64} = -3 \quad \text{A1}$$

(ii) $\log_{25} 5$

$$5 = 25^{1/2} \quad \text{M1}$$

$$\log_{25} 5 = \frac{1}{2} \quad \text{A1}$$

(iii) $\log_2 \sqrt{32}$

$$\sqrt{32} = 2^{5/2} \quad \text{M1}$$

$$\log_2 \sqrt{32} = \frac{5}{2} \quad \text{A1}$$

(b) Let $c = \log_3 m$. Write each expression in terms of c . [6]

(i) $\log_3 m^2 = 2 \log_3 m$ M1

$$= 2c \quad \text{A1}$$

(ii) $\log_3 27m = \log_3 27 + \log_3 m$ M1

$$= 3 + c \quad \text{A1}$$

(iii) $\log_9 m = \frac{\log_3 m}{\log_3 9}$ M1

$$= \frac{c}{2} \quad \text{A1}$$

2. [Maximum mark: 7]

Solve $\log_2(x - 3) = \log_4(x + 9)$.

Solution:

Change of base: $\log_4(x + 9) = \frac{1}{2} \log_2(x + 9)$ M1

$$2 \log_2(x - 3) = \log_2(x + 9) \quad \text{A1}$$

$$\log_2((x-3)^2) = \log_2(x+9)$$

$$\text{Equating arguments: } (x-3)^2 = x+9 \quad \text{M1}$$

$$x^2 - 6x + 9 = x + 9$$

$$x^2 - 7x = 0 \quad \text{A1}$$

$$x(x-7) = 0$$

$$x = 0 \text{ or } x = 7 \quad \text{A1}$$

$$\text{Domain requires } x-3 > 0, \text{ so } x > 3. \text{ Reject } x = 0. \quad \text{R1}$$

$$x = 7 \quad \text{A1}$$

3. [Maximum mark: 15]

(a) Series is geometric. [5]

(i) Show that $p = \pm \frac{1}{\sqrt{3}}$.

Solution:

$$\text{Common ratio: } \frac{p \ln x}{\ln x} = \frac{(1/3) \ln x}{p \ln x} \quad \text{M1}$$

$$p = \frac{1}{3p} \implies p^2 = \frac{1}{3} \quad \text{A1}$$

$$p = \pm \frac{1}{\sqrt{3}} \quad \text{AG}$$

(ii) Given $p > 0$ and $S_\infty = 3 + \sqrt{3}$, find x .

Solution:

$$r = \frac{1}{\sqrt{3}}, \quad a = \ln x$$

$$S_\infty = \frac{\ln x}{1 - \frac{1}{\sqrt{3}}} = \frac{\sqrt{3} \ln x}{\sqrt{3} - 1} = \frac{\sqrt{3}(\sqrt{3} + 1) \ln x}{2} = \frac{(3 + \sqrt{3}) \ln x}{2} \quad \text{M1}$$

$$\frac{(3 + \sqrt{3}) \ln x}{2} = 3 + \sqrt{3}$$

$$\ln x = 2 \quad \text{A1}$$

$$x = e^2 \quad \text{A1}$$

(b) Series is arithmetic with common difference d . [10]

(i) Show that $p = \frac{2}{3}$.

Solution:

$$p \ln x - \ln x = \frac{1}{3} \ln x - p \ln x \quad \text{M1}$$

$$(p - 1) \ln x = \left(\frac{1}{3} - p \right) \ln x$$

$$\text{Since } \ln x \neq 0: \quad 2p = \frac{4}{3} \quad \text{A1}$$

$$p = \frac{2}{3} \quad \text{AG}$$

(ii) Write down d in the form $k \ln x$.

Solution:

$$d = \left(\frac{2}{3} - 1 \right) \ln x = -\frac{1}{3} \ln x \quad \text{A1}$$

(iii) Sum of first n terms is $-3 \ln x$. Find n .

Solution:

$$S_n = \frac{n}{2} [2a + (n - 1)d] \quad \text{M1}$$

Substituting $a = \ln x$ and $d = -\frac{1}{3} \ln x$:

$$S_n = \frac{n}{2} \left[2 \ln x - \frac{n - 1}{3} \ln x \right] \quad \text{A1}$$

$$= \frac{n \ln x}{6} (7 - n) \quad \text{A1}$$

Set equal to $-3 \ln x$ (and divide by $\ln x > 0$): M1

$$\frac{n(7 - n)}{6} = -3$$

$$n^2 - 7n - 18 = 0 \quad \text{A1}$$

$$(n - 9)(n + 2) = 0 \quad \text{A1}$$

$$n = 9 \quad \text{A1}$$

Reject $n = -2$ since $n \in \mathbb{Z}^+$.

4. [Maximum mark: 14]

Let $g(x) = p^x + q$, for $x, p, q \in \mathbb{R}$, $p > 1$. The point $A(0, a)$ lies on the graph of g . Let $f(x) = g^{-1}(x)$.

(a) Write down the coordinates of B . [2]

Solution:

$$g(0) = p^0 + q = 1 + q, \text{ so } a = q + 1 \quad \text{A1}$$

$$B \text{ is the reflection of } A(0, a) \text{ in } y = x, \text{ so } B = (a, 0) \quad \text{A1}$$

(b) Given $f'(a) = \frac{1}{\ln p}$, find the equation of L_1 in terms of x , p and q . [5]

Solution:

$$L_1 \text{ passes through } B(a, 0) \text{ with gradient } \frac{1}{\ln p} \quad \text{M1}$$

$$y - 0 = \frac{1}{\ln p}(x - a) \quad \text{A1}$$

$$\text{Substituting } a = q + 1: \quad \text{M1}$$

$$y = \frac{1}{\ln p}(x - q - 1) \quad \text{A1A1}$$

FT: award A1A1 for the final equation using the candidate's a from part (a).

(c) Find the equation of L_1 in terms of x . [7]

Solution:

Finding p :

$$\text{Gradient of normal to } g \text{ at } A \text{ is } \frac{1}{\ln \frac{1}{3}} = -\frac{1}{\ln 3} \quad \text{M1}$$

$$(\ln p) \cdot \left(-\frac{1}{\ln 3}\right) = -1 \quad \text{A1}$$

$$\ln p = \ln 3 \implies p = 3 \quad \text{A1}$$

Finding q :

$$L_2 \text{ passes through } (-2, -2):$$

$$-2 = (\ln 3)(-2) + q + 1 \quad \text{A1}$$

$$q = -3 + 2 \ln 3 \quad \text{A1}$$

Finding L_1 :

$$a = q + 1 = -2 + 2 \ln 3 \quad \text{A1}$$

$$L_1: y = \frac{1}{\ln 3}(x - 2 \ln 3 + 2) \quad \text{A1}$$

5. [Maximum mark: 16]

Note: parts (b)–(e) require a GDC; values quoted to 3 s.f.

(a) Initial displacement. [2]

$$s(0) = 3(0 + 2) \cos 0 = 3 \cdot 2 \cdot 1 \quad \mathbf{M1}$$

$$s(0) = 6 \text{ metres} \quad \mathbf{A1}$$

(b) Velocity at $t = 2$. [2]

$$v(t) = s'(t) = 3 \cos t - 3(t + 2) \sin t \quad \mathbf{M1}$$

$$v(2) = 3 \cos 2 - 12 \sin 2 \approx -12.2 \text{ m/s} \quad \mathbf{A1}$$

(c) Intervals when the particle is moving away from P . [5]

Moving away from P means $|s|$ is increasing, i.e. s and v have the same sign.

$$\text{Zeros of } s: \cos t = 0 \Rightarrow t = \frac{\pi}{2}, \frac{3\pi}{2} \quad \mathbf{A1}$$

$$\text{Zeros of } v \text{ (GDC): } t \approx 0.402, 3.33, 6.41 \quad \mathbf{M1 A1}$$

Checking signs of s and v on each subinterval: $\mathbf{A1}$

Interval	sign of s	sign of v
$0 \leq t < 0.402$	+	+
$0.402 < t < \pi/2$	+	–
$\pi/2 < t < 3.33$	–	–
$3.33 < t < 3\pi/2$	–	+
$3\pi/2 < t < 6.41$	+	+
$6.41 < t < 6.8$	+	–

Moving away (same sign): $\mathbf{A1}$

$$0 \leq t < 0.402, \quad \frac{\pi}{2} < t < 3.33, \quad \frac{3\pi}{2} < t < 6.41$$

(d) Find b and c where acceleration is zero. [4]

$$a(t) = v'(t) = -6 \sin t - 3(t + 2) \cos t \quad \mathbf{M1}$$

Solving $a(t) = 0$ on $[0, 6.8]$ (GDC): $\mathbf{M1}$

$$b \approx 2.03 \quad \mathbf{A1}$$

$$c \approx 4.99 \quad \mathbf{A1}$$

(e) Total distance for $b \leq t \leq c$. [3]

$$\text{On } (b, c), v = 0 \text{ once at } t \approx 3.33. \text{ Total distance} = |s(3.33) - s(b)| + |s(c) - s(3.33)| \quad \mathbf{M1}$$

$$s(2.03) \approx -5.38, \quad s(3.33) \approx -15.7, \quad s(4.99) \approx 5.73 \quad \mathbf{A1}$$

$$\text{Total distance} \approx 10.3 + 21.4 \approx 31.7 \text{ m} \quad \mathbf{A1}$$